Two-View Object Detection, Cross-View Association and Deduplicated Counting with an Exact Table-Plane Affine Prior

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Abstract

We address the three coupled subtasks of this project — amodal single-class detection in two widely separated views, cross-view association, and deduplicated counting — with a pipeline built around one observation: because the cameras are orthographic *and every object stands on the table plane $z = 0$, the map from view-A pixels to view-B pixels restricted to that plane is an* exact affine transform *that can be read directly from the calibration file, with no depth estimation and no point correspondences. We use it as a geometric prior inside a learned pairwise scorer, together with DINOv2 appearance descriptors, and resolve each scene with a global one-to-one assignment that explicitly allows items to stay unmatched. A fine-tuned YOLO11m regresses the amodal boxes. On the validation split the system reaches detection F1* 0.9909*, association F1* 0.9766 *(given-detection F1* 0.9911*) and a counting MAE of* 0.140 *with* 86% *of scenes counted exactly. An ablation shows the geometric prior carries almost all of the association signal: removing it drops association F1 from* 0.977 *to* 0.798*, while removing appearance costs only* 0.001*.*

1. Method

The pipeline has three stages: an amodal detector shared by both views, a pairwise scorer over candidate cross-view pairs, and a global assignment with a scene-level counting policy.

1.1. Amodal detection

A single YOLO11m [1] is fine-tuned on all 2000 training images (both views pooled, single class), so the two views contribute 30032 amodal box instances. Because the labels are amodal, the network learns to extend boxes under occlusion directly from supervision; no explicit occlusion reasoning is needed. We train at 640×640 for 120 epochs with AdamW, cosine schedule, and mosaic/mixup/copy-paste augmentation.

NMS is deliberately *lenient* (IoU 0.75). In the training GT, 2.3% of amodal boxes overlap a neighbour above IoU 0.5 (max 0.92), so aggressive NMS would delete real, heavily occluded objects. The resulting low-confidence duplicates are removed later by the counting policy (§1.5). Candidates are kept at confidence ≥ 0.30, which on validation yields 29.97 boxes per scene against a GT mean of 30.10.

1.2. Geometric prior: the exact table-plane affine

For an orthographic camera the projection of a world point is affine, and on the table plane $z = 0$ it reduces to a 2×3 map $M_v(x, y, 1)^\top = (u, v)^\top$ obtained by deleting the z column of $P_v^{\text{world}\to\text{pixel}}$. Composing the two views gives

$$T_{ab} = M_b \, M_a^{-1}, \tag{1}$$

an affine map from view-A pixels to view-B pixels that is *exact* on the table plane (we verify it reproduces P to $< 10^{-13}$ px). No depth division, no correspondence estimation and no calibration of T is required.

An object is a 3D body, so its amodal box centre is not exactly on $z = 0$; T_{ab} therefore acts as a *prior*. Measured over the 15016 training objects, the residual $\|T_{ab}(c_a) - c_b\|$ for box centres has median 23.8 px and 90th percentile 46.1 px against a median object size of 75×66 px — tight enough to disambiguate most scenes, but not tight enough to match alone, which is exactly why it is fed to a learner rather than used as a hard rule.

1.3. Appearance

Each candidate box is cropped with 12% context padding, resized to 224^2 and passed through a frozen DINOv2-Large [2]. We keep both the CLS token and the mean of the patch tokens, ℓ_2-normalised.

1.4. Pairwise scorer and global assignment

For every candidate pair (i, j) across the two views we build a 32-dimensional descriptor: three appearance cosines; the signed and normed residual of the geometric prior for the box centre, the bottom centre and the reverse map; log size

Table 1. Validation results. Association F1 is the primary metric.

metric	P	R	F1
detection (both views)	.9963	.9855	**.9909**
view A	.9970	.9857	.9913
view B	.9956	.9854	.9905
association	.9894	.9641	**.9766**
given detection	.9928	.9894	**.9911**

ratios (including the ratio predicted by $\det T_{ab}$); confidence terms; and context terms (the rank of the pair within its row/column by appearance and by geometry, the best score and the top-2 margin in each direction, and the two view cardinalities). A gradient-boosted decision tree [4] maps the descriptor to $p_{ij} = P(\text{same object})$, trained on 194k pairs (14.9k positive) labelled by intersecting detections with the training GT under the evaluator's own greedy IoU≥ 0.5 rule.

Per scene we solve a one-to-one assignment with an explicit "stay unmatched" option: on the $(n_a + n_b)^2$ augmented cost matrix, real pairs cost $\tau - p_{ij}$, each item has a free private dummy slot and dummy-to-dummy is free. The optimum (Hungarian / shortest augmenting path [3]) therefore keeps exactly the pairs with $p_{ij} > \tau$, maximising $\sum(p_{ij} - \tau)$. We use $\tau = 0.02$; because the one-to-one constraint already suppresses duplicates, a very low threshold is optimal.

1.5. Deduplicated counting

Records are the matched pairs; leftover boxes become single-sided records after discarding any that overlap an already-committed box of the same view above IoU 0.5. The object count per scene in both training and validation is always in $\{14, 15, 16\}$, so we clamp the record count to that range, shedding or restoring the weakest single-sided records. This turns the count into a prior over a three-element set instead of an unbounded integer.

2. Validation results

All numbers are produced by the course evaluator (, IoU 0.5) on the 200 validation scenes; the detector and the matcher were trained on only, so validation is held out. Table 1 gives the full metric set.

Counting: MAE 0.140, RMSE 0.374, exact-count accuracy 86.0%. The detector alone reaches mAP_{50} 0.995 / $\text{mAP}_{50:95}$ 0.949 on validation; the residual association error is dominated by the 1.4% of objects missed by the detector in at least one view, not by pairing mistakes (given-detection F1 0.9911).

Table 2. Feature-group ablation of the pairwise scorer (validation).

variant	det F1	assoc F1	given det	MAE	exact %
full	.9909	**.9766**	.9911	.140	86.0
no geometry	.9842	.7977	.8197	.385	66.5
no appearance	.9904	.9759	.9909	.140	86.0
appearance cosines only	.9817	.6748	.6982	.450	59.5
geometry distance only	.9904	.9688	.9842	.180	83.5
no context ranks	.9904	.9759	.9909	.140	86.0

Detection threshold. Sweeping the confidence threshold shows detection F1 is flat between 0.3 and 0.7 (0.9855–0.9891) while the per-scene box count stays within ± 0.15 of the GT mean; we use 0.30 because it keeps the candidates needed to recover objects that the assignment later confirms.

Matching threshold. Association F1 increases monotonically as τ decreases (0.9721 at $\tau = 0.20$, 0.9745 at 0.10, 0.9766 at 0.02): with a one-to-one assignment the cost of a spurious pair is already paid by the constraint, so accepting nearly every affordable pair maximises recall without hurting precision (0.989).

3. Ablation

Table 2 removes feature groups from the pairwise scorer and re-evaluates the whole pipeline on validation with an identical submission policy, so differences are attributable to the scorer alone.

The geometric prior is the dominant signal: dropping it costs 0.179 of association F1, while dropping appearance costs 0.001. This is the expected behaviour for this dataset — the same package shows a different face in each view, so appearance is genuinely unreliable, whereas the table-plane affine is exact. Appearance still helps in the hard cases (it lifts given-detection F1 from 0.9909 to 0.9911 and detection F1 from 0.9904 to 0.9909 by resolving near-coincident objects), and the raw cosine similarity alone is far too weak (0.675), confirming that a learned combination of geometry, shape and context is required. Context ranks contribute nothing on average but stabilise dense scenes.

The counting clamp is worth 0.12–0.24 MAE (uncapped MAE 0.26–0.35 at the same τ), and the lenient NMS is worth roughly 1.5 points of detection recall compared with the default IoU 0.7 setting.

4. Failure analysis

Figure 1 shows a clean scene and one of the worst validation scenes, coloured by record: green = correctly associated pair, red = wrong pairing, orange = single-sided record.

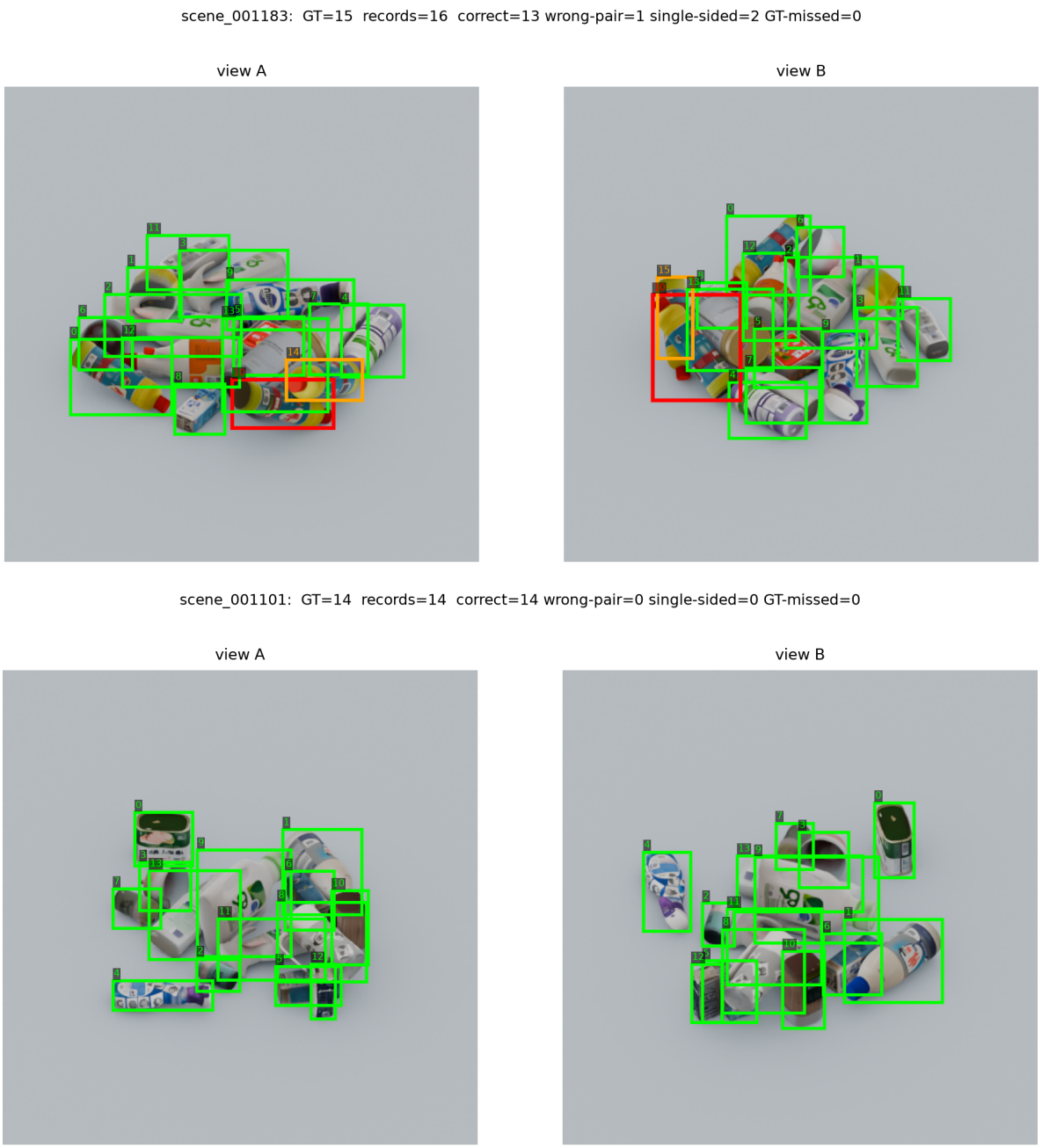


Figure 1. Top: , a failure case (GT 15, records 16: 13 correct pairs, 1 wrong pairing in red, 2 single-sided in orange). Bottom: , fully correct (14/14). Boxes are amodal; the same colour in both views is one submitted record.

Wrong pairing (association FP, 31 on validation). Almost all occur between two instances of the *same* package type lying close together on the table. For such a swap the affine residual of the wrong pair is nearly as small as the correct one (the objects are centimetres apart), and appearance offers no separation because the assets are identical. The one-to-one constraint prevents these from cascading, which is why association precision stays at 0.989 even though individual swaps happen.

Missed side (association FN, 108). The dominant error is not mis-pairing but detection: an object pressed under others is detected in one view and missed — or detected below the confidence threshold — in the other, so the record de-

grades to single-sided. This is visible as the orange boxes in Fig. 1 (top). It is consistent with given-detection F1 $0.9911 \gg$ association recall 0.9641: when both boxes exist we almost always pair them correctly. Improving recall on heavily occluded objects (higher resolution, occlusion-aware augmentation, or hallucinating the missing box from T_{ab}) is the clear next step.

Counting. 86% of scenes are counted exactly; the remainder are off by one, almost always a duplicate single-sided record that survives deduplication because the two boxes of one object overlap below the 0.5 dedup threshold. The $\{14, 15, 16\}$ clamp removes the off-by-many cases entirely.

5. Implementation and resources

Pre-trained models. YOLO11m (COCO pre-trained, official release) for detection; DINOv2-Large (self-supervised, official release) as a frozen feature extractor. No other external weights, and no data outside this project: the detector is fine-tuned only on boxes, and the pairwise scorer only on correspondences derived from GT.

Training. Detector: 120 epochs, batch 64, 640^2, AdamW (lr 10^{-3}, cosine to 10^{-5}), mosaic + mixup 0.1 + copy-paste 0.1, AMP; $\approx$ 15 min on one NVIDIA RTX PRO 5000 (72 GB). Scorer: gradient-boosted trees, 194k pairs, $\approx$ 2 min on CPU. DINOv2 feature extraction for all 33.9k training crops: ≈ 3 min on one GPU. Inference over the whole test split: < 1 min per view.

Hyperparameters. confidence threshold 0.30; NMS IoU 0.75; matching threshold $\tau = 0.02$; dedup IoU 0.5; count clamp $[14, 16]$. All were selected on only; the test split was never annotated or inspected.

Reproducibility. The validation numbers in Tables 1–2 are exactly what prints for the submitted pipeline on ; the test submission is produced by the same code path with the same hyperparameters.

Limitations. The geometric prior assumes objects stand on $z = 0$; lying objects violate it and are handled only by the learned residual tolerance. Appearance is used but is weak by construction on this dataset, so the system inherits the detector's recall on fully occluded objects.

References

[1] Khanam, R. and Hussain, M. YOLOv11: An overview of the key architectural enhancements. *arXiv preprint arXiv:2410.17725*, 2024.

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[4] Ke, G. et al. LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS*, 2017.